Stat 6390.001: High Dimensional Probability and Statistics, Fall 2026 **Topic 20**

Random matrices and covariance estimation

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Introduction

A random matrix has random variables as its entries. Write

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1d} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{nd} \end{pmatrix} = \begin{pmatrix} x_1^{\mathrm{T}} \\ \vdots \\ x_n^{\mathrm{T}} \end{pmatrix} \in \mathbb{R}^{n\times d}, \qquad x_i = (x_{i1}, \ldots, x_{id})^{\mathrm{T}}.$$

For covariance estimation, suppose $x_1, \ldots, x_n$ are i.i.d. with mean zero. Their common covariance matrix is

$$\begin{aligned} \operatorname{cov}(x_1) &= \mathbb{E}\big[(x_1 - \mathbb{E}[x_1])(x_1 - \mathbb{E}[x_1])^{\mathrm{T}}\big] \\ &= \mathbb{E}[x_1 x_1^{\mathrm{T}}] - \mathbb{E}[x_1]\big(\mathbb{E}[x_1]\big)^{\mathrm{T}} = \mathbb{E}[x_1 x_1^{\mathrm{T}}]. \end{aligned}$$

The matrix product gives

$$\begin{aligned} \frac{1}{n}\mathbf{X}^{\mathrm{T}}\mathbf{X} &= \frac{1}{n}\begin{pmatrix} x_1 & \cdots & x_n \end{pmatrix}\begin{pmatrix} x_1^{\mathrm{T}} \\ \vdots \\ x_n^{\mathrm{T}} \end{pmatrix} \\ &= \frac{1}{n}\big(x_1 x_1^{\mathrm{T}} + \cdots + x_n x_n^{\mathrm{T}}\big) = \frac{1}{n}\sum_{i=1}^{n} x_i x_i^{\mathrm{T}}. \end{aligned}$$

This matrix estimates $\operatorname{cov}(x_1)$. We consider the regimes $d < n$, $d \asymp n$, and $d \gg n$.

Asymptotic results

Wigner semicircle law. Suppose $\mathbf{W}_n = (w_{ij})_{i,j=1}^n$ is real symmetric ($w_{ji} = w_{ij}$), with independent upper-triangular entries of mean zero and unit variance. Let $\lambda_i(\mathbf{A}_n)$ be the ith eigenvalue of the normalized matrix $\mathbf{A}_n = \mathbf{W}_n/\sqrt{n}$. The empirical distribution of these eigenvalues is

$$\mu_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{\lambda_i(\mathbf{A}_n)},$$

which places mass $1/n$ at each eigenvalue. Under the hypotheses of the semicircle law, $\mu_n \Rightarrow \mu_{\mathrm{sc}}$ as $n \to \infty$, where μ_{sc} has density

$$f_{\mathrm{sc}}(x) = \frac{\sqrt{4 - x^2}}{2\pi}\,\mathbb{I}[|x| < 2].$$

Although there are $n(n+1)/2$ independent entries, the eigenvalues of $\mathbf{W}_n$ are on the scale of $\sqrt{n}$. The trace of $\mathbf{W}_n^2$ gives the intuition.

$$\begin{aligned}\operatorname{tr}(\mathbf{W}_n^2) &= \sum_{i=1}^n \sum_{j=1}^n w_{ij} w_{ji} = \sum_{i=1}^n \sum_{j=1}^n w_{ij}^2 \\ &= \sum_{k=1}^n \lambda_k(\mathbf{W}_n^2) = \sum_{k=1}^n \big(\lambda_k(\mathbf{W}_n)\big)^2.\end{aligned}$$

Since $\mathbb{E}[w_{ij}] = 0$, we have $\operatorname{var}(w_{ij}) = \mathbb{E}[w_{ij}^2] - (\mathbb{E}[w_{ij}])^2 = \mathbb{E}[w_{ij}^2] = 1$, so

$$\mathbb{E}\left[\sum_{i=1}^n \sum_{j=1}^n w_{ij}^2\right] = n^2.$$

This suggests $\sum_{k=1}^n (\lambda_k(\mathbf{W}_n))^2 \approx n^2$, so $|\lambda_k(\mathbf{W}_n)| \approx \sqrt{n}$ if the eigenvalues have comparable magnitudes.

Remark. The eigenvalues are dependent. The eigenvectors are delocalized, meaning that their components are spread across the coordinates.

Marčenko–Pastur law. Suppose $\mathbf{X} \in \mathbb{R}^{n \times d}$ has rows $x_1^{\mathrm{T}}, \ldots, x_n^{\mathrm{T}}$, with $\mathbb{E}[x_{ij}] = 0$ and $\operatorname{var}(x_{ij}) = 1$. The sample covariance matrix is

$$\widehat{\mathbf{\Sigma}} = \frac{1}{n} \mathbf{X}^{\mathrm{T}} \mathbf{X} \in \mathbb{R}^{d \times d}.$$

Assume $d, n \to \infty$ with $d/n \to \gamma \in (0, \infty)$. In the Marčenko–Pastur setting, the empirical distribution of the eigenvalues of $\widehat{\mathbf{\Sigma}}$ converges as follows.

$$\mu_n = \frac{1}{n} \sum_{i=1}^d \delta_{\lambda_i(\widehat{\mathbf{\Sigma}})} \;\Rightarrow\; \mu_{\mathrm{MP},\gamma}.$$

The density of the continuous part of $\mu_{\mathrm{MP},\gamma}$ is

$$f_\gamma(x) = \frac{\sqrt{(b-x)(x-a)}}{2\pi\gamma x}\, \mathbb{I}[a \le x \le b], \qquad a = (1-\sqrt{\gamma})^2, \qquad b = (1+\sqrt{\gamma})^2.$$

If $\gamma > 1$, the distribution also has a point mass at zero of size $1 - 1/\gamma$, and $\int_a^b f_\gamma(x)\, dx = 1/\gamma$.

When $\gamma > 1$, we eventually have $d > n$, so $\operatorname{rank}(\mathbf{X}^{\mathrm{T}}\mathbf{X}) = \operatorname{rank}(\mathbf{X}) \le n < d$. Consequently, $\mathbf{X}^{\mathrm{T}}\mathbf{X}$ has at least $d - n$ zero eigenvalues, and

$$\frac{d-n}{d} = 1 - \frac{n}{d} \;\longrightarrow\; 1 - \frac{1}{\gamma}.$$

For example, when $\gamma = 1$, we have $a = 0$ and $b = 4$. Even when $\mathbf{\Sigma} = \mathbf{I}_d$, the sample eigenvalues can range asymptotically from 0 to 4.

Under suitable moment assumptions, when $0 < \gamma \le 1$, we have

$$\lambda_{\min}(\widehat{\mathbf{\Sigma}}) \longrightarrow (1-\sqrt{\gamma})^2, \qquad \lambda_{\max}(\widehat{\mathbf{\Sigma}}) \longrightarrow (1+\sqrt{\gamma})^2.$$

Thus, in the identity-covariance case, $\|\widehat{\mathbf{\Sigma}} - \mathbf{I}_d\|_2$ fails to converge to zero in probability. In particular, $v^{\mathrm{T}}(\widehat{\mathbf{\Sigma}} - \mathbf{I}_d)v$ fails to converge to zero in probability when v is a unit eigenvector corresponding to $\lambda_{\max}(\widehat{\mathbf{\Sigma}})$.

We seek non-asymptotic probability bounds for finite n and d. We start with Gaussian random matrices, then consider sub-Gaussian ensembles and ensembles with milder tail conditions.

6.1 Some preliminaries

6.1.1 Notation and basic facts

Given a square matrix $\mathbf{A} \in \mathbb{R}^{n\times n}$, an eigenvalue λ and a corresponding nonzero eigenvector v satisfy $\mathbf{A}v = \lambda v$. If $\mathbf{A}$ is real and symmetric, it has the decomposition

$$\mathbf{A} = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{\mathrm{T}} = \begin{pmatrix} v_1 & \cdots & v_n \end{pmatrix} \begin{pmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{pmatrix} \begin{pmatrix} v_1^{\mathrm{T}} \\ \vdots \\ v_n^{\mathrm{T}} \end{pmatrix}.$$

For $\mathbf{A} \in \mathbb{R}^{n\times m}$ with $n \geq m$, the ith singular value is the square root of the ith eigenvalue of $\mathbf{A}^{\mathrm{T}}\mathbf{A}$. Thus,

$$\sigma_i(\mathbf{A}) = \sqrt{\lambda_i(\mathbf{A}^{\mathrm{T}}\mathbf{A})}.$$

The singular values are nonnegative. The singular value decomposition is $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^{\mathrm{T}}$, with

$$\mathbf{A}v_j = \sigma_j(\mathbf{A})u_j, \qquad \mathbf{A}^{\mathrm{T}}u_j = \sigma_j(\mathbf{A})v_j.$$

We order the singular values as

$$\sigma_{\max}(\mathbf{A}) = \sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq \sigma_m(\mathbf{A}) = \sigma_{\min}(\mathbf{A}) \geq 0.$$

The largest and smallest singular values have the variational characterizations

$$\sigma_{\max}(\mathbf{A}) = \max_{v\in\mathbb{S}^{d-1}} \|\mathbf{A}v\|_2, \qquad \sigma_{\min}(\mathbf{A}) = \min_{v\in\mathbb{S}^{d-1}} \|\mathbf{A}v\|_2. \tag{6.1}$$

Here $\mathbb{S}^{d-1} \coloneqq \{v \in \mathbb{R}^d \mid \|v\|_2 = 1\}$ is the Euclidean unit sphere in $\mathbb{R}^d$, and $\|\mathbf{A}v\|_2^2 = v^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}\mathbf{A}v$. The matrix ℓ_2-operator norm, also called the spectral norm, is

$$|||\mathbf{A}|||_2 \coloneqq \sup_{\|v\|_2=1} \|\mathbf{A}v\|_2 = \sigma_{\max}(\mathbf{A}).$$

Other texts also write this norm as $\|\mathbf{A}\|_{\mathrm{op}}$ or $\|\mathbf{A}\|_2$.

Since covariance matrices are symmetric, we focus on the set $\mathcal{S}^{d\times d} \coloneqq \{\mathbf{Q} \in \mathbb{R}^{d\times d} \mid \mathbf{Q} = \mathbf{Q}^{\mathrm{T}}\}$ of symmetric matrices and the subset of positive semidefinite matrices

$$\mathcal{S}_+^{d\times d} \coloneqq \{\mathbf{Q} \in \mathcal{S}^{d\times d} \mid \mathbf{Q} \succeq 0\}. \tag{6.2}$$

Any $\mathbf{Q} \in \mathcal{S}^{d\times d}$ is diagonalizable by an orthogonal transformation. In particular, $\mathbf{Q} = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{\mathrm{T}}$ with $\mathbf{P}^{\mathrm{T}}\mathbf{P} = \mathbf{P}\mathbf{P}^{\mathrm{T}} = \mathbf{I}_d$. We denote its vector of eigenvalues by $\gamma(\mathbf{Q}) \in \mathbb{R}^d$, ordered as

$$\gamma_{\max}(\mathbf{Q}) = \gamma_1(\mathbf{Q}) \geq \gamma_2(\mathbf{Q}) \geq \cdots \geq \gamma_d(\mathbf{Q}) = \gamma_{\min}(\mathbf{Q}).$$

The matrix $\mathbf{Q}$ is positive semidefinite, written $\mathbf{Q} \succeq 0$, if and only if $\gamma_{\min}(\mathbf{Q}) \geq 0$.

The Rayleigh–Ritz variational characterization gives

$$\gamma_{\max}(\mathbf{Q}) = \max_{v\in\mathbb{S}^{d-1}} v^{\mathrm{T}}\mathbf{Q}v, \qquad \gamma_{\min}(\mathbf{Q}) = \min_{v\in\mathbb{S}^{d-1}} v^{\mathrm{T}}\mathbf{Q}v. \tag{6.3}$$

For symmetric $\mathbf{Q}$, $\mathbf{Q}^{\mathrm{T}}\mathbf{Q} = \mathbf{Q}^2$ and

$$\mathbf{Q}^2 = \mathbf{P}\mathbf{\Lambda}\mathbf{P}^{\mathrm{T}}\mathbf{P}\mathbf{\Lambda}\mathbf{P}^{\mathrm{T}} = \mathbf{P}\mathbf{\Lambda}^2\mathbf{P}^{\mathrm{T}}.$$

Consequently, the operator norm can be written as

$$\begin{aligned} |||\mathbf{Q}|||_2 &= \max_{v \in \mathbb{S}^{d-1}} \|\mathbf{Q}v\|_2 = \max_{v \in \mathbb{S}^{d-1}} \left(v^{\mathrm{T}} \mathbf{Q}^{\mathrm{T}} \mathbf{Q} v\right)^{1/2} \\ &= \left[\gamma_{\max}(\mathbf{Q}^2)\right]^{1/2} = \max\left\{\gamma_{\max}(\mathbf{Q}), \, |\gamma_{\min}(\mathbf{Q})|\right\}. \end{aligned} \tag{6.4a}$$

It follows that

$$|||\mathbf{Q}|||_2 = \max_{v \in \mathbb{S}^{d-1}} \left|v^{\mathrm{T}} \mathbf{Q} v\right|. \tag{6.4b}$$

Finally, for $\mathbf{A} \in \mathbb{R}^{n \times m}$ with $n \geq m$, let m- dimensional symmetric $\mathbf{R} := \mathbf{A}^{\mathrm{T}}\mathbf{A}$. Then we have relationship

$$\gamma_j(\mathbf{R}) = \left(\sigma_j(\mathbf{A})\right)^2, \qquad j = 1, \ldots, m.$$

6.1.2 Set-up of covariance estimation

Let $\{x_1, \ldots, x_n\}$ be n i.i.d. samples from a distribution in $\mathbb{R}^d$ with mean zero. We use a lowercase x_i for a random vector and uppercase $\mathbf{X}$ for the random matrix. The covariance is

$$\mathbf{\Sigma} = \operatorname{cov}(x_1) = \mathbb{E}\left[(x_1 - \mathbb{E}[x_1])(x_1 - \mathbb{E}[x_1])^{\mathrm{T}}\right].$$

The sample covariance matrix estimates $\mathbf{\Sigma}$ and is given by

$$\widehat{\mathbf{\Sigma}} := \frac{1}{n} \sum_{i=1}^{n} x_i x_i^{\mathrm{T}}. \tag{6.5}$$

Since each x_i has mean zero, $\mathbb{E}[x_i x_i^{\mathrm{T}}] = \mathbf{\Sigma}$ and $\mathbb{E}[\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}] = 0$. Thus $\widehat{\mathbf{\Sigma}}$ is unbiased.

Our goal is to bound the estimation error in the ℓ_2-operator norm. The sample quadratic form is

$$v^{\mathrm{T}} \widehat{\mathbf{\Sigma}} v = v^{\mathrm{T}} \left(\frac{1}{n} \sum_{i=1}^{n} x_i x_i^{\mathrm{T}}\right) v = \frac{1}{n} \sum_{i=1}^{n} (v^{\mathrm{T}} x_i)(x_i^{\mathrm{T}} v) = \frac{1}{n} \sum_{i=1}^{n} \langle x_i, v \rangle^2.$$

By (6.4b), the bound $|||\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}|||_2 \leq \epsilon$ is equivalent to

$$\max_{v \in \mathbb{S}^{d-1}} \left| \frac{1}{n} \sum_{i=1}^{n} \langle x_i, v \rangle^2 - v^{\mathrm{T}} \mathbf{\Sigma} v \right| \leq \epsilon. \tag{6.6}$$

Since $v^{\mathrm{T}} \mathbf{\Sigma} v = \mathbb{E}[\langle x_i, v \rangle^2]$, this is also equivalent to

$$\max_{v \in \mathbb{S}^{d-1}} \left| \frac{1}{n} \sum_{i=1}^{n} \left(\langle x_i, v \rangle^2 - \mathbb{E}[\langle x_i, v \rangle^2]\right) \right| \leq \epsilon.$$

Thus controlling the operator norm error amounts to establishing a uniform law of large numbers for $f_v(x) = \langle x, v \rangle^2$, indexed by $v \in \mathbb{S}^{d-1}$ (see Chapter 4).

Control in operator norm also controls the eigenvalue errors uniformly. By Weyl's theorem,

$$\max_{j=1,\ldots,d} \left|\gamma_j(\widehat{\mathbf{\Sigma}}) - \gamma_j(\mathbf{\Sigma})\right| \leq |||\widehat{\mathbf{\Sigma}} - \mathbf{\Sigma}|||_2. \tag{6.7}$$

An analogous eigenvector guarantee requires control on gaps between adjacent eigenvalues, for example $\gamma_j(\mathbf{\Sigma}) > \gamma_{j+1}(\mathbf{\Sigma}) + \Delta$ for a positive gap Δ (see Chapter 8).

Finally, let $\{\sigma_j(\mathbf{X})\}_{j=1}^{\min\{n,d\}}$ denote the singular values of the random matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$ with rows x_i^{T}. Then

$$\widehat{\mathbf{\Sigma}} = \frac{1}{n} \sum_{i=1}^{n} x_i x_i^{\mathrm{T}} = \frac{1}{n} \mathbf{X}^{\mathrm{T}} \mathbf{X} = \left(\frac{\mathbf{X}}{\sqrt{n}}\right)^{\mathrm{T}} \left(\frac{\mathbf{X}}{\sqrt{n}}\right).$$

The eigenvalues of $\widehat{\mathbf{\Sigma}}$ are the squared singular values of $\mathbf{X}/\sqrt{n}$, padded with zeros if $n < d$.

6.2 Wishart matrices and their behavior

Suppose $x_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \mathbf{\Sigma})$ for $i = 1, \dots, n$. We say that $\mathbf{X} \in \mathbb{R}^{n \times d}$, with ith row x_i^{T}, is drawn from the $\mathbf{\Sigma}$-Gaussian ensemble. For the sample covariance $\widehat{\mathbf{\Sigma}} = \mathbf{X}^{\mathrm{T}}\mathbf{X}/n$, the Wishart relation is

$$n\widehat{\mathbf{\Sigma}} \sim W_d(n, \mathbf{\Sigma}).$$

When $d = 1$ and the variance is σ^2, this gives $W_1(n, \sigma^2)$.

Theorem 6.1. *Let $\mathbf{X} \in \mathbb{R}^{n \times d}$ be drawn from the $\mathbf{\Sigma}$-Gaussian ensemble, with $\mathbf{\Sigma} \neq 0$. For all $\delta > 0$, the maximum singular value satisfies*

$$\mathbb{P}\left[\frac{\sigma_{\max}(\mathbf{X})}{\sqrt{n}} \geq \gamma_{\max}(\sqrt{\mathbf{\Sigma}})(1+\delta) + \sqrt{\frac{\operatorname{tr}(\mathbf{\Sigma})}{n}}\right] \leq e^{-n\delta^2/2}. \tag{6.8}$$

For $n \geq d$, the minimum singular value satisfies

$$\mathbb{P}\left[\frac{\sigma_{\min}(\mathbf{X})}{\sqrt{n}} \leq \gamma_{\min}(\sqrt{\mathbf{\Sigma}})(1-\delta) - \sqrt{\frac{\operatorname{tr}(\mathbf{\Sigma})}{n}}\right] \leq e^{-n\delta^2/2}. \tag{6.9}$$

For $n < d$, the variational minimum $\min_{v \in \mathbb{S}^{d-1}} \|\mathbf{X}v\|_2$ is zero.

Example 6.2 (Operator norm bounds for the standard Gaussian ensemble)**.** Let $\mathbf{W} \in \mathbb{R}^{n \times d}$ have rows w_i^{T}, where $w_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, \mathbf{I}_d)$. Suppose $n \geq d$. Let A and B be the events in the brackets of (6.8) and (6.9) for $\mathbf{X} = \mathbf{W}$ and $\mathbf{\Sigma} = \mathbf{I}_d$. On $A^c \cap B^c$ we have

$$\frac{\sigma_{\max}(\mathbf{W})}{\sqrt{n}} \leq 1 + \delta + \sqrt{\frac{d}{n}}, \qquad \frac{\sigma_{\min}(\mathbf{W})}{\sqrt{n}} \geq 1 - \delta - \sqrt{\frac{d}{n}}. \tag{6.10}$$

By Theorem 6.1, both $\mathbb{P}(A^c)$ and $\mathbb{P}(B^c)$ are at least $1 - e^{-n\delta^2/2}$. Hence

$$\begin{aligned}\mathbb{P}(A^c \cap B^c) &= \mathbb{P}(A^c) + \mathbb{P}(B^c) - \mathbb{P}(A^c \cup B^c) \\ &\geq \mathbb{P}(A^c) + \mathbb{P}(B^c) - 1 \geq 1 - 2e^{-n\delta^2/2}.\end{aligned}$$

The bounds in (6.10) imply

$$\left\| \frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W} - \mathbf{I}_d \right\|_2 \leq 2\epsilon + \epsilon^2, \qquad \epsilon = \sqrt{\frac{d}{n}} + \delta, \tag{6.11}$$

with the same probability lower bound.

To see this, write $M = \sigma_{\max}(\mathbf{W})/\sqrt{n}$ and $m = \sigma_{\min}(\mathbf{W})/\sqrt{n}$. Then $-\epsilon \leq m - 1 \leq M - 1 \leq \epsilon$, so each singular value satisfies

$$1 - \epsilon \leq \frac{\sigma_j(\mathbf{W})}{\sqrt{n}} \leq 1 + \epsilon.$$

Using nonnegativity of the singular values, we can square the bounds as

$$\big(\max\{0, 1-\epsilon\}\big)^2 \leq \left(\frac{\sigma_j(\mathbf{W})}{\sqrt{n}}\right)^2 \leq (1+\epsilon)^2. \tag{*}$$

The eigenvalues of $\mathbf{W}^{\mathrm{T}}\mathbf{W}/n$ are precisely these squared singular values. If $\mathbf{W}^{\mathrm{T}}\mathbf{W}/n = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{\mathrm{T}}$, then

$$\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W} - \mathbf{I}_d = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^{\mathrm{T}} - \mathbf{V}\mathbf{I}_d\mathbf{V}^{\mathrm{T}} = \mathbf{V}(\mathbf{\Lambda} - \mathbf{I}_d)\mathbf{V}^{\mathrm{T}}.$$

Thus subtracting $\mathbf{I}_d$ shifts each eigenvalue by -1, so

$$\begin{aligned}\left\|\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\right\|_2 &= \max_{1\le j\le d}\left|\gamma_j\left(\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\right)\right| \\ &= \max_{1\le j\le d}\left|\gamma_j\left(\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}\right)-1\right| \\ &= \max_{1\le j\le d}\left|\left(\frac{\sigma_j(\mathbf{W})}{\sqrt{n}}\right)^2-1\right| \\ &\le (1+\epsilon)^2-1 = 2\epsilon+\epsilon^2.\end{aligned}$$

The inequality follows from (*), since $1-(\max\{0,1-\epsilon\})^2 \le (1+\epsilon)^2-1$.

If $d/n \to 0$, we can choose $\delta_n \to 0$ with $n\delta_n^2 \to \infty$, for instance $\delta_n = \sqrt{\log(n)/n}$. Then $\epsilon \to 0$ and the failure probability bound $2e^{-n\delta_n^2/2}$ tends to zero. Thus $\widehat{\mathbf{\Sigma}} = \mathbf{W}^{\mathrm{T}}\mathbf{W}/n$ is consistent in operator norm, with

$$\|\widehat{\mathbf{\Sigma}} - \mathbf{I}_d\|_2 \xrightarrow{\mathbb{P}} 0.$$

Example 6.3 (Gaussian covariance estimation)**.** Let the rows of $\mathbf{X}$ be x_i^{T}, with $x_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\mathbf{\Sigma})$. Writing $x_i = \sqrt{\mathbf{\Sigma}}\, w_i$, where $w_i \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0,\mathbf{I}_d)$, gives $\mathbf{X} = \mathbf{W}\sqrt{\mathbf{\Sigma}}$. Hence

$$\begin{aligned}\left\|\frac{1}{n}\mathbf{X}^{\mathrm{T}}\mathbf{X}-\mathbf{\Sigma}\right\|_2 &= \left\|\sqrt{\mathbf{\Sigma}}\Big(\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\Big)\sqrt{\mathbf{\Sigma}}\right\|_2 \\ &\le \|\sqrt{\mathbf{\Sigma}}\|_2^2\left\|\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\right\|_2 \\ &= \|\mathbf{\Sigma}\|_2\left\|\frac{1}{n}\mathbf{W}^{\mathrm{T}}\mathbf{W}-\mathbf{I}_d\right\|_2.\end{aligned}$$

The inequality uses submultiplicativity of the operator norm, which follows from $\|\mathbf{A}\mathbf{B}v\|_2 \le \|\mathbf{A}\|_2\|\mathbf{B}v\|_2$. The last equality uses positive semidefiniteness of $\mathbf{\Sigma}$.

Applying (6.11) for $n \ge d$ gives, for $\mathbf{\Sigma} \ne 0$ and $\delta > 0$,

$$\frac{\|\widehat{\mathbf{\Sigma}}-\mathbf{\Sigma}\|_2}{\|\mathbf{\Sigma}\|_2} \le 2\sqrt{\frac{d}{n}} + 2\delta + \left(\sqrt{\frac{d}{n}}+\delta\right)^2. \tag{6.12}$$

This bound holds with probability at least $1-2e^{-n\delta^2/2}$. The relative error converges to zero in probability as $d/n \to 0$.

We now consider additional structure through control of the eigenvalues of $\mathbf{\Sigma}$.

Example 6.4 (Faster rates under trace constraints)**.** Let $\{\gamma_j(\mathbf{\Sigma})\}_{j=1}^d$ be the ordered eigenvalues of a nonzero covariance matrix $\mathbf{\Sigma}$, with $\gamma_1(\mathbf{\Sigma})$ its largest eigenvalue. Consider

$$\frac{\operatorname{tr}(\mathbf{\Sigma})}{|||\mathbf{\Sigma}|||_2} = \frac{\sum_{j=1}^d \gamma_j(\mathbf{\Sigma})}{\gamma_1(\mathbf{\Sigma})} \leq C, \tag{6.13}$$

where C is independent of dimension. One possible bound is $C = \operatorname{rank}(\mathbf{\Sigma})$.

A matrix can be full rank and still have relatively fast eigenvalue decay. Consider the Schatten q-balls

$$\mathbb{B}_q(R_q) \coloneqq \left\{ \mathbf{\Sigma} \in \mathcal{S}^{d\times d} \;\middle|\; \sum_{j=1}^d |\gamma_j(\mathbf{\Sigma})|^q \leq R_q \right\}, \tag{6.14}$$

where $q \in [0,1]$ and the radius is $R_q > 0$. At $q = 0$, the sum counts the nonzero eigenvalues.

Restrict attention to matrices with eigenvalues in $[-1,1]$. For $q_1 < q_2$,

$$\sum_{j=1}^d |\gamma_j(\mathbf{\Sigma})|^{q_2} \leq \sum_{j=1}^d |\gamma_j(\mathbf{\Sigma})|^{q_1}.$$

Thus, at a fixed radius R, the restricted sets are nested as $\mathbb{B}_{q_1}(R) \subseteq \mathbb{B}_{q_2}(R)$. The smallest set has $q = 0$ and the largest has $q = 1$. Under this eigenvalue restriction, a nonzero covariance matrix $\mathbf{\Sigma} \in \mathbb{B}_q(R_q)$ satisfies (6.13) with $C = R_q/(\gamma_1(\mathbf{\Sigma}))^q$.

For any matrix class satisfying (6.13), Theorem 6.1 gives, with high probability,

$$\frac{\sigma_{\max}(\mathbf{X})}{\sqrt{n}} \leq \gamma_{\max}(\sqrt{\mathbf{\Sigma}})\left(1 + \delta + \sqrt{\frac{C}{n}}\right). \tag{6.15}$$

Comparing this with (6.10) for $\mathbf{\Sigma} = \mathbf{I}_d$ shows that C plays the role of the effective dimension.